NHANES 1999 & 2011

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Reading Files: NHANES 1999

```
library(haven)

#Be sure to replace the file path with the appropriate file path on your own computer.

setwd("/Users/adriana/Library/CloudStorage/OneDrive-Emory/0 TB_DM/NHANES/NHANES 1999-2000")

#reading the datasets and putting them into one file
file00 <- c("DEMO.XPT",
            "TBQ.xpt",
            "DIQ.XPT",
            "LAB10AM.xpt",
            "LAB10.xpt",
            "LAB16.xpt",
            "TB.xpt",
            "LAB13.XPT",
            "LAB13AM.XPT",
            "PH.XPT",
            "BPX.XPT",
            "BMX.XPT"
            )

#assgning a name for each dataset
for(i in 1:length(file00)) {
  oname = paste0("df", i)
  assign(oname, read_xpt(file00[i]))
}

#merging the data set
data_nhanes_1999 <- merge(df1, df2,  by ="SEQN", all = T)
data_nhanes_1999 <- merge(data_nhanes_1999, df3,  by ="SEQN", all = T)
data_nhanes_1999 <- merge(data_nhanes_1999, df4,  by ="SEQN", all = T)
data_nhanes_1999 <- merge(data_nhanes_1999, df5,  by ="SEQN", all = T)
data_nhanes_1999 <- merge(data_nhanes_1999, df6,  by ="SEQN", all = T)
data_nhanes_1999 <- merge(data_nhanes_1999, df7,  by ="SEQN", all = T)
data_nhanes_1999 <- merge(data_nhanes_1999, df8,  by ="SEQN", all = T)
data_nhanes_1999 <- merge(data_nhanes_1999, df9,  by ="SEQN", all = T)
data_nhanes_1999 <- merge(data_nhanes_1999, df10,  by ="SEQN", all = T)
data_nhanes_1999 <- merge(data_nhanes_1999, df11,  by ="SEQN", all = T)
data_nhanes_1999 <- merge(data_nhanes_1999, df12,  by ="SEQN", all = T)
```

```
# cleaning
rm(df1, df2, df3, df4, df5,
   df6, df7, df8, df9, df10, df11, df12, file00, i, oname)
```

Cleaning: NHANES 1999

```
attach(data_nhanes_1999)

#creating new data frame which only has the variables of interest
df_nhanes_1999 <- data.frame(
  id_study = 'NHANES_1999_2000',
  id       = SEQN,

  psu1     = as.numeric(SDMVPSU),
  psu2     = NA,

  stratum1 = as.numeric(SDMVSTRA),
  stratum2 = NA,

  psu_weight1 = NA,
  psu_weight2 = NA,
  psu_weight3 = NA,

  # demographics
  sex     = DMDHRGND, #RIAGENDR
  age     = as.numeric(RIDAGEYR),
  race    = RIDRETH1,
  edu_adult = DMDEDUC2,
  edu_years = NA,
  edu_child = DMDEDUC3,
  ethnicity = RIDRETH2,

  #diabetes
  self_db      = DIQ010,
  age_told_db  = DIQ040Q,
  drug_insu_db = DIQ050,
  drug_pill_db = DIQ070,
  family_hx_db = NA,

  #labs
  fasting_minutes = as.numeric(PHAFSTMN),
  fasting_hours   = as.numeric(PHAFSTHR),
  fasting_8h      = NA,
  fasting_12h     = NA,
  fglu         = as.numeric(LBXGLU), units_fglu  = 'mg/dL',
  hba1c        = as.numeric(LBXGH), units_hba1c = '%',
  pogtt          = NA, units_pogtt   = NA,
  insulin      = as.numeric(LBXIN), units_insulin = 'uU/mL',

  creatinine_urine = as.numeric(URXUCR), creatinine_urine_units = 'mg/dL',
```

```
  total_cholesterol        = as.numeric(LBXTC), units_cholesterol = 'mg/dL',
  ldl                      = as.numeric(LBDLDL), units_ldl = 'mg/dL',

  #tb
  tb_history = TBQ040,
  age_told_tb  = NA,
  tb_treatment = TBQ050,
  tb_skin_test = TBQ010,
  tb_skin_test_result = TBQ020,
  tb_medication_use = TBQ030,

  # Measurements
  weight                   = as.numeric(BMXWT), units_weight = 'kg',
  height                   = as.numeric(BMXHT), units_height = 'cm',
  bmi                      = as.numeric(BMXBMI), units_bmi = 'KG/M**2',
  waist                    = as.numeric(BMXWAIST), units_waist = 'cm',            # just 1 measure
  hip                      = NA, units_hip = NA,
  # These will be converted into blood_pressure_sys and blood_pressure_diast later
  blood_pressure_sys2       = as.numeric(BPXSY2), units_sbp = 'mmHg',
  blood_pressure_sys3       = as.numeric(BPXSY3), units_sbp = 'mmHg',
  blood_pressure_diast2     = as.numeric(BPXDI2),
  blood_pressure_diast3     = as.numeric(BPXDI3)
)

detach(data_nhanes_1999)
```

Reading Files: NHANES 2011

```
library(haven)

# Be sure to replace the file path with the appropriate file path on your own computer.

setwd("/Users/adriana/Library/CloudStorage/OneDrive-Emory/0 TB_DM/NHANES/NHANES 2011-2012")

#reading the datasets and putting them into one file
file_list <- c("DEMO_G.xpt",
               "TBQ_G.xpt",
               "DIQ_G.xpt",
               "GLU_G.xpt",
               "GHB_G.xpt",
               "ALB_CR_G.xpt",
               "TB_G.xpt",
               "FASTQX_G.XPT",
               "TCHOL_G.XPT",
               "TRIGLY_G.XPT",
               "BMX_G.XPT",
               "BPX_G.XPT"
               )

#assgning a name for each dataset
for(i in 1:length(file_list)) {
```

```
  oname = paste0("df", i)
  assign(oname, read_xpt(file_list[i]))
}

#merging the data set
data_nhanes_2011 <- merge(df1, df2,  by ="SEQN", all = T)
data_nhanes_2011 <- merge(data_nhanes_2011, df3,  by ="SEQN", all = T)
data_nhanes_2011 <- merge(data_nhanes_2011, df4,  by ="SEQN", all = T)
data_nhanes_2011 <- merge(data_nhanes_2011, df5,  by ="SEQN", all = T)
data_nhanes_2011 <- merge(data_nhanes_2011, df6,  by ="SEQN", all = T)
data_nhanes_2011 <- merge(data_nhanes_2011, df7,  by ="SEQN", all = T)
data_nhanes_2011 <- merge(data_nhanes_2011, df8,  by ="SEQN", all = T)
data_nhanes_2011 <- merge(data_nhanes_2011, df9,  by ="SEQN", all = T)
data_nhanes_2011 <- merge(data_nhanes_2011, df10,  by ="SEQN", all = T)
data_nhanes_2011 <- merge(data_nhanes_2011, df11,  by ="SEQN", all = T)
data_nhanes_2011 <- merge(data_nhanes_2011, df12,  by ="SEQN", all = T)

# cleaning
rm(df1, df2, df3, df4, df5,
   df6, df7, df8, df9, df10, df11, df12, file_list, i, oname)
```

Cleaning NHANES 2011

```
# cleaning

attach(data_nhanes_2011)

#creating new data frame which only has the variables of interest
df_nhanes_2011 <- data.frame(
  id_study = 'NHANES_2011_2012',
  id       = SEQN,

  psu1     = as.numeric(SDMVPSU),
  psu2     = NA,

  stratum1 = as.numeric(SDMVSTRA),
  stratum2 = NA,

  psu_weight1 = NA,
  psu_weight2 = NA,
  psu_weight3 = NA,

  # demographics
  sex      = RIAGENDR,
  age      = as.numeric(RIDAGEYR),
  race     = RIDRETH1,
  edu_adult = DMDEDUC2,
  edu_years = NA,
  edu_child = DMDEDUC3,
  ethnicity = RIDRETH3,
```

```
  #diabetes
  self_db      = DIQ010,
  age_told_db  = NA, #object 'DIQ040Q' not found
  drug_insu_db = DIQ050,
  drug_pill_db = DIQ070,
  family_hx_db = DIQ175A,

  #labs
  fasting_minutes = as.numeric(PHAFSTMN.y),
  fasting_hours   = as.numeric(PHAFSTHR.y),
  fasting_8h      = NA,
  fasting_12h     = NA,
  fglu         = as.numeric(LBXGLU), units_fglu  = 'mg/dL',
  hba1c        = as.numeric(LBXGH), units_hba1c = '%',
  pogtt          = NA, units_pogtt   = 'NA',
  insulin      = as.numeric(LBXIN), units_insulin = 'uU/mL',

  creatinine_urine = as.numeric(URXUCR), creatinine_urine_units = 'mg/dL',

  total_cholesterol        = as.numeric(LBXTC), units_cholesterol = 'mg/dL',
  ldl                      = as.numeric(LBDLDL), units_ldl = 'mg/dL',

  #tb
  tb_history = TBQ040,
  age_told_tb  = NA,
  tb_treatment = TBQ050,
  tb_skin_test = TBQ015A,
  tb_skin_test_result = TBQ022,
  tb_medication_use = TBQ030,

  # Measurements
  weight                   = as.numeric(BMXWT), units_weight = 'lbs',
  height                   = as.numeric(BMXHT), units_height = 'cm',
  bmi                      = as.numeric(BMXBMI), units_bmi = 'KG/M**2',
  waist                    = as.numeric(BMXWAIST), units_waist = 'cm',
  hip                      = NA,                   units_hip = 'cm',
  blood_pressure_sys2      = as.numeric(BPXSY2), units_sbp = 'mmHg',
  blood_pressure_diast2    = as.numeric(BPXDI2), units_dbp = 'mmHg',
  blood_pressure_sys3      = as.numeric(BPXSY3),
  blood_pressure_diast3    = as.numeric(BPXDI3)
)

detach(data_nhanes_2011)
```

Re-coding both datasets

```
recode_nhanes <- function(df) {
  df %>%
    mutate(

      # sex
```

```
    sex = case_when(
      sex == 1 ~ 1, #men
      sex == 2 ~ 2, #women
      TRUE ~ NA_real_
    ),

    # yes/no variables
    across(
      c(self_db, drug_insu_db, drug_pill_db,
        tb_history, tb_treatment, tb_medication_use,
        tb_skin_test, tb_skin_test_result),
      ~ case_when(
        . == 1 ~ 1,
        . == 2 ~ 0,
        . %in% c(7, 9) ~ 9,
        TRUE ~ NA_real_
      )
    ),

    # family history
    family_hx_db = case_when(
      family_hx_db == 10 ~ 1,
      TRUE ~ NA_real_
    ),

    # race
    race = case_when(
      race == 4 ~ 0,
      race == 3 ~ 1,
      race %in% c(1, 2) ~ 2,
      race == 5 ~ 3,
      race == 8 ~ 9,                # DK
      TRUE ~ NA_real_
    ),

    # ethnicity
    ethnicity = case_when(
      ethnicity %in% c(1, 2, 4) ~ 1,
      ethnicity %in% c(3, 5) ~ 2,
      TRUE ~ NA_real_
    ),

    # education
    edu_adult = case_when(
      edu_adult %in% c(1, 2) ~ 1,
      edu_adult == 3 ~ 2,
      edu_adult == 4 ~ 3,
      edu_adult == 5 ~ 4,
      edu_adult %in% c(7, 9) ~ 9,
      TRUE ~ 9
    )

  )
```

```
}

# Re coding both datasets (1999 and 2011)
df_nhanes_2011 <- recode_nhanes(df_nhanes_2011)
df_nhanes_1999 <- recode_nhanes(df_nhanes_1999)
```

Diabetes definitions

```
# Cleaning fasting glucose:
df_nhanes_2011 <- df_nhanes_2011 %>%      # NA increased: 6723 -> 6903
  mutate(
    fglu = case_when(
      fasting_hours < 8 ~ NA_real_,
      TRUE ~ fglu
    )
  )

df_nhanes_1999 <- df_nhanes_1999 %>%      # NA increased: 6931 -> 7182
  mutate(
    fglu = case_when(
      fasting_hours < 8 ~ NA_real_,
      TRUE ~ fglu
    )
  )


# Cleaning diabetes-lab variables:

library(dplyr)

df_nhanes_2011 <- df_nhanes_2011 %>%
  mutate(
    fglu  = ifelse(fglu  >= 36 & fglu  <= 540, fglu,  NA),      # Plausible range: 36 - 540 mg/dl
    hba1c = ifelse(hba1c >= 3  & hba1c <= 18,  hba1c, NA)       # Plausible range: 3 - 18 %
  )

df_nhanes_1999 <- df_nhanes_1999 %>%
  mutate(
    fglu  = ifelse(fglu  >= 36 & fglu  <= 540, fglu,  NA),      # Plausible range: 36 - 540 mg/dl
    hba1c = ifelse(hba1c >= 3  & hba1c <= 18,  hba1c, NA)       # Plausible range: 3 - 18 %
  )



# Diabetes: Self reported / labs (fasting glucose, HbA1c, oral glucose tolerance test)
create_self_lab_dm <- function(dm) {
  dm %>%
    mutate(
      self_lab_dm = case_when(
        self_db == 1 | fglu >= 126 | hba1c >= 6.5 | pogtt >= 200 ~ 1,
        !is.na(self_db) | !is.na(fglu) | !is.na(hba1c) | !is.na(pogtt) ~ 0,
```

```
        is.na(self_db) & is.na(fglu) & is.na(hba1c) & is.na(pogtt) ~ NA_real_
      )
    )
}

df_nhanes_1999 <- create_self_lab_dm(df_nhanes_1999)
df_nhanes_2011 <- create_self_lab_dm(df_nhanes_2011)
```

Generating same SBP and DBP as other datasets

```
# Average of blood pressure measurement (systolic and diastolic #2 + #3 / 2)
process_bp <- function(df) {
  df$blood_pressure_sys   <- (df$blood_pressure_sys2 + df$blood_pressure_sys3) / 2
  df$blood_pressure_diast <- (df$blood_pressure_diast2 + df$blood_pressure_diast3) / 2

  df <- subset(df, select = -c(blood_pressure_sys2,
                               blood_pressure_sys3,
                               blood_pressure_diast2,
                               blood_pressure_diast3))
  return(df)
}

df_nhanes_1999 <- process_bp(df_nhanes_1999)
df_nhanes_2011 <- process_bp(df_nhanes_2011)

# Summary of numeric variables - 1999
summary(df_nhanes_1999$blood_pressure_sys)
summary(df_nhanes_1999$blood_pressure_diast)

library(dplyr)

df_nhanes_1999 <- df_nhanes_1999 %>%
  mutate(
    blood_pressure_sys = ifelse(
      blood_pressure_sys >= 70 & blood_pressure_sys <= 270,
      blood_pressure_sys, NA
    ),
    blood_pressure_diast = ifelse(
      blood_pressure_diast >= 30 & blood_pressure_diast <= 150,
      blood_pressure_diast, NA
    )
  )


# Summary of numeric variables - 2011

summary(df_nhanes_2011$blood_pressure_sys)
summary(df_nhanes_2011$blood_pressure_diast)


df_nhanes_2011 <- df_nhanes_2011 %>%
```

```
  mutate(
    blood_pressure_sys = ifelse(
      blood_pressure_sys >= 70 & blood_pressure_sys <= 270,
      blood_pressure_sys, NA
    ),
    blood_pressure_diast = ifelse(
      blood_pressure_diast >= 30 & blood_pressure_diast <= 150,
      blood_pressure_diast, NA
    )
  )
```

Numeric variables

```
# Summary of numeric variables - 1999
summary(df_nhanes_1999$weight)
summary(df_nhanes_1999$height)
summary(df_nhanes_1999$bmi)
summary(df_nhanes_1999$waist)
summary(df_nhanes_1999$hip)
summary(df_nhanes_1999$ldl)
summary(df_nhanes_1999$total_cholesterol)


library(dplyr)

df_nhanes_1999 <- df_nhanes_1999 %>%
  mutate(
    weight = ifelse(weight >= 26.45 & weight <= 661.39, weight, NA),        # Plausible range: 12-300 k
    height = ifelse(height >= 100 & height <= 250, height, NA),             # Plausible range: 100 - 25
    bmi = ifelse(bmi >= 10 & bmi <= 80, bmi, NA),                           # Plausible range: 10-80 kg
    waist = ifelse(waist >= 30 & waist <= 300, waist, NA),                  # Plausible range: 30 - 300
    hip = ifelse(hip >= 40 & hip <= 300, hip, NA),                          # Plausible range: 40 - 300
    ldl = ifelse(ldl >= 19.33 & ldl <= 386.65, ldl, NA),                    # Plausible range: 0.5 - 10
    total_cholesterol = ifelse(
      total_cholesterol >= 1.75 & total_cholesterol <= 20,                  # Plausible range: 1.75 - 2
      total_cholesterol, NA
    )
  )


# Summary of numeric variables - 2011
summary(df_nhanes_2011$weight)
summary(df_nhanes_2011$height)
summary(df_nhanes_2011$bmi)
summary(df_nhanes_2011$waist)
summary(df_nhanes_2011$hip)
summary(df_nhanes_2011$ldl)
summary(df_nhanes_2011$total_cholesterol)

df_nhanes_2011 <- df_nhanes_2011 %>%
  mutate(
```

```
    weight = ifelse(weight >= 26.45 & weight <= 661.39, weight, NA),        # Plausible range: 12-300 k
    height = ifelse(height >= 100 & height <= 250, height, NA),             # Plausible range: 100 - 25
    bmi = ifelse(bmi >= 10 & bmi <= 80, bmi, NA),                           # Plausible range: 10-80 kg
    waist = ifelse(waist >= 30 & waist <= 300, waist, NA),                  # Plausible range: 30 - 300
    hip = ifelse(hip >= 40 & hip <= 300, hip, NA),                          # Plausible range: 40 - 300
    ldl = ifelse(ldl >= 19.33 & ldl <= 386.65, ldl, NA),                    # Plausible range: 0.5 - 10
    total_cholesterol = ifelse(
      total_cholesterol >= 1.75 & total_cholesterol <= 20,                  # Plausible range: 1.75 - 2
      total_cholesterol, NA
    )
  )
```

Labeling both datasets

```
label_nhanes <- function(label) {
  label %>%
    mutate(
      # Sex
      sex = factor(sex,
                   levels = c(1, 2),
                   labels = c("Men", "Women")),

      # TB variables
      tb_history = factor(tb_history,
                          levels = c(1, 0, 9),
                          labels = c("Yes", "No", "Refused/DK")),
      tb_treatment = factor(tb_treatment,
                            levels = c(1, 0, 9),
                            labels = c("Yes", "No", "Refused/DK")),
      tb_skin_test = factor(tb_skin_test,
                            levels = c(1, 0, 9),
                            labels = c("Yes", "No", "Refused/DK")),
      tb_skin_test_result = factor(tb_skin_test_result,
                                   levels = c(1, 0, 9),
                                   labels = c("Positive", "Negative", "Refused/DK")),
      tb_medication_use = factor(tb_medication_use,
                                 levels = c(1, 0, 9),
                                 labels = c("Yes", "No", "Refused/DK")),

      # Diabetes self-report + labs
      self_db = factor(self_db,
                       levels = c(1, 0, 9),
                       labels = c("Yes", "No", "Refused/DK")),
      drug_insu_db = factor(drug_insu_db,
                            levels = c(1, 0, 9),
                            labels = c("Yes", "No", "Refused/DK")),
      drug_pill_db = factor(drug_pill_db,
                            levels = c(1, 0, 9),
                            labels = c("Yes", "No", "Refused/DK")),
      self_lab_dm = factor(self_lab_dm,
                           levels = c(1, 0),
```

```
                          labels = c("Diabetes", "No Diabetes")),

      # Family history
      family_hx_db = factor(family_hx_db,
                            levels = c(1, 0),
                            labels = c("Yes", "No")),

      # Race/Ethnicity
      race = factor(race,
                    levels = c(0, 1, 2, 3, 9),
                    labels = c("Non-Hispanic Black",
                               "Non-Hispanic White",
                               "Mexican/Other Hispanic",
                               "Other/Multiracial",
                               "Missing")),
      ethnicity = factor(ethnicity,
                         levels = c(1, 2, 9),
                         labels = c("Non-Hispanic", "Hispanic", "Missing")),

      # Education
      edu_adult = factor(edu_adult,
                         levels = c(1, 2, 3, 4, 9),
                         labels = c("< High School",
                                    "High School Grad",
                                    "Some College/AA",
                                    "College+",
                                    "Refused/DK"))
    )
}

df_nhanes_1999 <- label_nhanes(df_nhanes_1999)
df_nhanes_2011 <- label_nhanes(df_nhanes_2011)
```

Exploratory data analysis

```
###==== 1999 - 2000 =======
#TB (n=35) 1=yes -> 1 // 2=no -> 0 // 7=refused // 9=DK // .=NA
table(df_nhanes_1999$tb_history)

#DM (n=630) 1=yes 2=no 3=borderline 7=refused 9=DK .=NA
table(df_nhanes_1999$self_db)       # DM self reported (n=489)
table(df_nhanes_1999$self_lab_dm)   # DM self reported + Labs (n=630)

#DM and TB (n=3)
table(df_nhanes_1999$tb_history, df_nhanes_1999$self_db) # DM and TB self reported (n=1)
table(df_nhanes_1999$tb_history, df_nhanes_1999$self_lab_dm) # DM self reported&Lab and TB self reporte


###==== 2011 - 2012 =======
#TB (n=44)
table(df_nhanes_2011$tb_history)
```

```r
#DM (n=708)
table(df_nhanes_2011$self_db)        # DM self reported (n=708)
table(df_nhanes_2011$self_lab_dm)    # DM self reported + Labs (n=946)


#DM and TB (n=5)
table(df_nhanes_2011$tb_history, df_nhanes_2011$self_db) # DM and TB self reported (n=4)
table(df_nhanes_2011$tb_history, df_nhanes_2011$self_lab_dm) # DM self reported&Lab and TB self reporte
```

New DM and TB variable

```r
# For 1999
df_nhanes_1999 <- df_nhanes_1999 %>%
  mutate(
    tb_dm = case_when(
      self_lab_dm == "Diabetes" & tb_history == "Yes" ~ 1,  # both DM and TB
      TRUE ~ 0                                               # everything else
    )
  )

# For 2011
df_nhanes_2011 <- df_nhanes_2011 %>%
  mutate(
    tb_dm = case_when(
      self_lab_dm == "Diabetes" & tb_history == "Yes" ~ 1,
      TRUE ~ 0
    )
  )

# Labeling both datasets
label_tb_dm <- function(tb_dm) {
  tb_dm %>%
    mutate(
      # diabetes & tuberculosis
      tb_dm = factor(tb_dm,
                     levels = c(0, 1),
                     labels = c("No", "Yes"))
    )
}

df_nhanes_1999 <- label_tb_dm(df_nhanes_1999)
df_nhanes_2011 <- label_tb_dm(df_nhanes_2011)

#Checking
table(df_nhanes_1999$tb_dm)
table(df_nhanes_2011$tb_dm)
```

Preliminary results:

NHANES 1999

```
library(gtsummary)

# Select the variables you want to include
vars_to_summarize <- c(
  "age", "sex", "race", "self_db", "age_told_db", "drug_insu_db", "drug_pill_db",
  "self_lab_dm", "family_hx_db",
  "tb_history", "age_told_tb", "tb_treatment", "fasting_minutes", "fglu", "hba1c", "total_cholesterol",

# Create table1
NHANES_1999_table <- df_nhanes_1999 %>%
  select(all_of(vars_to_summarize), tb_dm) %>%  # include the grouping variable
  tbl_summary(
    by = tb_dm,  # group by TB+DM
    type = list(
      age ~ "continuous",
      age_told_db ~ "continuous",
      age_told_tb ~ "continuous",
      fglu ~ "continuous",
      hba1c ~ "continuous",
      fasting_minutes ~ "continuous",
      total_cholesterol ~ "continuous",
      ldl ~ "continuous",
      weight ~ "continuous",
      height ~ "continuous",
      bmi ~ "continuous",
      waist ~ "continuous",
      hip ~ "continuous",
      blood_pressure_sys ~ "continuous",
      blood_pressure_diast ~ "continuous"
    ),
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} / {N} ({p}%)"
    ),
    label = var_label(df_nhanes_1999),
    missing = "no"
  )

NHANES_1999_table <- NHANES_1999_table %>%
  modify_header(label ~ "**Variables**") %>%
  modify_spanning_header(c(stat_1, stat_2) ~ "**Summary Statistics (NHANES 1999)**")

NHANES_1999_table
```

Preliminary results:

NHANES 2011

```
library(gtsummary)

NHANES_2011_table <- df_nhanes_2011 %>%
  select(all_of(vars_to_summarize), tb_dm) %>%  # include the grouping variable
  tbl_summary(
    by = tb_dm,  # group by TB+DM
    type = list(
      age ~ "continuous",
      age_told_db ~ "continuous",
      age_told_tb ~ "continuous",
      fglu ~ "continuous",
      hba1c ~ "continuous",
      fasting_minutes ~ "continuous",
      total_cholesterol ~ "continuous",
      ldl ~ "continuous",
      weight ~ "continuous",
      height ~ "continuous",
      bmi ~ "continuous",
      waist ~ "continuous",
      hip ~ "continuous",
      blood_pressure_sys ~ "continuous",
      blood_pressure_diast ~ "continuous"
    ),
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} / {N} ({p}%)"
    ),
    label = var_label(df_nhanes_1999),
    missing = "no"
  )

NHANES_2011_table <- NHANES_2011_table %>%
  modify_header(label ~ "**Variables**") %>%
  modify_spanning_header(c(stat_1, stat_2) ~ "**Summary Statistics (NHANES 2011)**")

NHANES_2011_table
```

Preliminary results:

NHANES 1999 AND 2011

```
library(dplyr)

df_nhanes_both <- bind_rows(
  df_nhanes_1999,
```

```
  df_nhanes_2011
)

# checking
table(df_nhanes_both$tb_dm) # n=455

# "Clean" dataset
df_nhanes_both <- df_nhanes_both %>%
  filter(tb_dm == "Yes")          # n=8
```

```
library(gtsummary)

NHANES_both_table <- df_nhanes_both %>%
  select(all_of(vars_to_summarize)) %>%   # remove tb_dm from the selection
  tbl_summary(
    type = list(
      age ~ "continuous",
      age_told_db ~ "continuous",
      age_told_tb ~ "continuous",
      fglu ~ "continuous",
      hba1c ~ "continuous",
      fasting_minutes ~ "continuous",
      total_cholesterol ~ "continuous",
      ldl ~ "continuous",
      weight ~ "continuous",
      height ~ "continuous",
      bmi ~ "continuous",
      waist ~ "continuous",
      hip ~ "continuous",
      blood_pressure_sys ~ "continuous",
      blood_pressure_diast ~ "continuous"
    ),
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} / {N} ({p}%)"
    ),
    label = var_label(df_nhanes_1999),
    missing = "no"
  ) %>%
  modify_header(label ~ "**Variables**") %>%
  modify_spanning_header(everything() ~ "**Summary Statistics (NHANES 1999 and 2011)**")

NHANES_both_table
```

Variables	Summary Statistics (NHANES 1999)	
	No N = 9,962[1]	**Yes** N = 3[1]
Age	30 (25)	69 (4)
Sex		
Men	5,594 / 9,950 (56%)	3 / 3 (100%)
Women	4,356 / 9,950 (44%)	0 / 3 (0%)
Race		
Non-Hispanic Black	2,227 / 9,962 (22%)	1 / 3 (33%)
Non-Hispanic White	3,367 / 9,962 (34%)	0 / 3 (0%)
Mexican/Other Hispanic	3,980 / 9,962 (40%)	2 / 3 (67%)
Other/Multiracial	388 / 9,962 (3.9%)	0 / 3 (0%)
Missing	0 / 9,962 (0%)	0 / 3 (0%)
Self-reported diabetes		
Yes	488 / 9,427 (5.2%)	1 / 3 (33%)
No	8,934 / 9,427 (95%)	2 / 3 (67%)
Refused/DK	5 / 9,427 (<0.1%)	0 / 3 (0%)
Age told DM	682 (7,935)	43 (NA)
Diabetes Mellitus on insulin		
Yes	137 / 6,646 (2.1%)	1 / 3 (33%)
No	6,507 / 6,646 (98%)	2 / 3 (67%)
Refused/DK	2 / 6,646 (<0.1%)	0 / 3 (0%)
Diabetes Mellitus on Oral med		
Yes	334 / 488 (68%)	0 / 1 (0%)
No	152 / 488 (31%)	1 / 1 (100%)
Refused/DK	2 / 488 (0.4%)	0 / 1 (0%)
DM self-reported and by lab def		
Diabetes	614 / 9,481 (6.5%)	3 / 3 (100%)
No Diabetes	8,867 / 9,481 (94%)	0 / 3 (0%)
Family history of DM	0 / 0 (NA%)	0 / 0 (NA%)
TB History ever		
Yes	32 / 9,485 (0.3%)	3 / 3 (100%)
No	9,401 / 9,485 (99%)	0 / 3 (0%)
Refused/DK	52 / 9,485 (0.5%)	0 / 3 (0%)
Age told TB	NA (NA)	NA (NA)
TB treatment ever		
Yes	25 / 32 (78%)	2 / 3 (67%)
No	6 / 32 (19%)	1 / 3 (33%)
Refused/DK	1 / 32 (3.1%)	0 / 3 (0%)
Fasting (minutes)	28 (17)	22 (24)
Fasting Glucose (mg/dl)	98 (27)	109 (NA)
HbA1c (%)	5.41 (1.00)	8.17 (1.81)
Total cholesterol (mg/dL)	NA (NA)	NA (NA)
LDL (mg/dL)	112 (35)	187 (NA)
Weight (lbs)	71 (22)	88 (18)
Height (cm)	159 (19)	165 (2)
BMI (kg/m2)	25 (7)	32 (7)
Waist (cm)	85 (20)	113 (15)
Hip (cm)	16NA (NA)	NA (NA)
Systolic blood pressure (mmHg)	119 (20)	127 (4)
Diastolic blood pressure (mmHg)	69 (12)	63 (13)

[1]Mean (SD); n / N (%)

Variables	Summary Statistics (NHANES 2011)	
	No N = 9,751[1]	**Yes** N = 5[1]
Age	31 (25)	58 (5)
Sex		
Men	4,854 / 9,751 (50%)	2 / 5 (40%)
Women	4,897 / 9,751 (50%)	3 / 5 (60%)
Race		
Non-Hispanic Black	2,680 / 9,751 (27%)	3 / 5 (60%)
Non-Hispanic White	2,973 / 9,751 (30%)	0 / 5 (0%)
Mexican/Other Hispanic	2,431 / 9,751 (25%)	0 / 5 (0%)
Other/Multiracial	1,667 / 9,751 (17%)	2 / 5 (40%)
Missing	0 / 9,751 (0%)	0 / 5 (0%)
Self-reported diabetes		
Yes	704 / 9,233 (7.6%)	4 / 5 (80%)
No	8,523 / 9,233 (92%)	1 / 5 (20%)
Refused/DK	6 / 9,233 (<0.1%)	0 / 5 (0%)
Age told DM	NA (NA)	NA (NA)
Diabetes Mellitus on insulin		
Yes	213 / 9,358 (2.3%)	0 / 5 (0%)
No	9,144 / 9,358 (98%)	5 / 5 (100%)
Refused/DK	1 / 9,358 (<0.1%)	0 / 5 (0%)
Diabetes Mellitus on Oral med		
Yes	543 / 1,074 (51%)	3 / 4 (75%)
No	528 / 1,074 (49%)	1 / 4 (25%)
Refused/DK	3 / 1,074 (0.3%)	0 / 4 (0%)
DM self-reported and by lab def		
Diabetes	931 / 9,348 (10.0%)	5 / 5 (100%)
No Diabetes	8,417 / 9,348 (90%)	0 / 5 (0%)
Family history of DM	1,164 / 1,164 (100%)	1 / 1 (100%)
TB History ever		
Yes	39 / 8,156 (0.5%)	5 / 5 (100%)
No	8,076 / 8,156 (99%)	0 / 5 (0%)
Refused/DK	41 / 8,156 (0.5%)	0 / 5 (0%)
Age told TB	NA (NA)	NA (NA)
TB treatment ever		
Yes	27 / 39 (69%)	3 / 5 (60%)
No	11 / 39 (28%)	2 / 5 (40%)
Refused/DK	1 / 39 (2.6%)	0 / 5 (0%)
Fasting (minutes)	29 (17)	23 (17)
Fasting Glucose (mg/dl)	105 (31)	126 (19)
HbA1c (%)	5.70 (1.06)	7.08 (1.60)
Total cholesterol (mg/dL)	NA (NA)	NA (NA)
LDL (mg/dL)	110 (35)	105 (39)
Weight (lbs)	73 (24)	81 (26)
Height (cm)	159 (20)	163 (13)
BMI (kg/m2)	25 (8)	30 (8)
Waist (cm)	86 (22)	106 (18)
Hip (cm)	17 NA (NA)	NA (NA)
Systolic blood pressure (mmHg)	118 (18)	146 (39)
Diastolic blood pressure (mmHg)	68 (13)	79 (8)

[1]Mean (SD); n / N (%)

Summary Statistics (NHANES 1999 and 2011)	
Variables	**N = 8**[1]
Age	61.9 (7.4)
Sex	
Men	5 / 8 (63%)
Women	3 / 8 (38%)
Race	
Non-Hispanic Black	4 / 8 (50%)
Non-Hispanic White	0 / 8 (0%)
Mexican/Other Hispanic	2 / 8 (25%)
Other/Multiracial	2 / 8 (25%)
Missing	0 / 8 (0%)
Self-reported diabetes	
Yes	5 / 8 (63%)
No	3 / 8 (38%)
Refused/DK	0 / 8 (0%)
Age told DM	43.0000 (NA)
Diabetes Mellitus on insulin	
Yes	1 / 8 (13%)
No	7 / 8 (88%)
Refused/DK	0 / 8 (0%)
Diabetes Mellitus on Oral med	
Yes	3 / 5 (60%)
No	2 / 5 (40%)
Refused/DK	0 / 5 (0%)
DM self-reported and by lab def	
Diabetes	8 / 8 (100%)
No Diabetes	0 / 8 (0%)
Family history of DM	1 / 1 (100%)
TB History ever	
Yes	8 / 8 (100%)
No	0 / 8 (0%)
Refused/DK	0 / 8 (0%)
Age told TB	NA (NA)
TB treatment ever	
Yes	5 / 8 (63%)
No	3 / 8 (38%)
Refused/DK	0 / 8 (0%)
Fasting (minutes)	23 (18)
Fasting Glucose (mg/dl)	122 (17)
HbA1c (%)	7.49 (1.65)
Total cholesterol (mg/dL)	NA (NA)
LDL (mg/dL)	125 (52)
Weight (lbs)	84 (22)
Height (cm)	164 (10)
BMI (kg/m2)	30.9 (6.9)
Waist (cm)	109 (16)
Hip (cm)	NA (NA)
Systolic blood pressure (mmHg)	136 (27)
Diastolic blood pressure (mmHg)	71 (13)
TB & DM	8 / 8 (100%)